Which training videos are worth keeping?

EgoScore · EgoVerse Data Optimization & Evaluation Suite, Track 1 · github.com/sahilmahendrakar/egoscore

THE PROBLEM

EgoVerse is a library of videos of people performing everyday tasks, recorded from a head-worn camera. Robots learn manipulation from them. Some of its clips are unusable — the hand tracking fails, or a recording covers a whole session rather than one task. Many others are near-duplicates that add training time without adding information.

HOW IT WORKS

1. **Remove clips that are unusable.**

   Eight checks on the recorded positions: frozen tracking, a hand jumping past 10 m/s, a clip over 3× its lab's median.

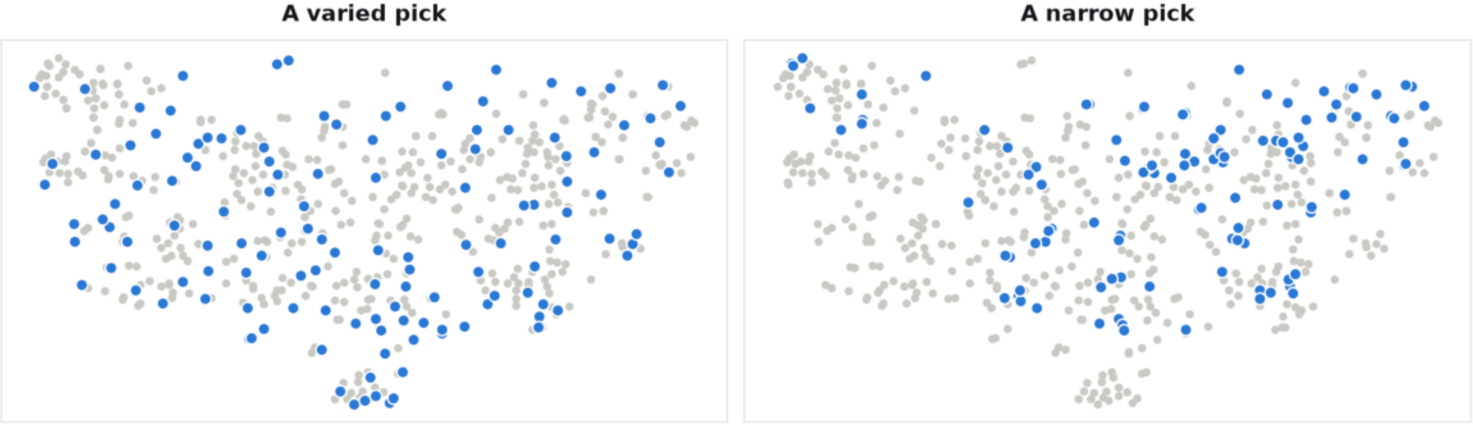
2. **Select a varied subset of the rest.**

   Summarise each clip as 41 numbers describing how the person moved, then choose clips unlike one another, not at random.

3. **Measure whether it helped.**

   Train an identical model on each subset; score it on clips from people and rooms absent from its training.

Both sides picked the same 130 videos out of 572



covers 97 of 190 person-and-room combinations
the model trained on it was 5% better

covers 20 of 190 person-and-room combinations
the model trained on it was 11% worse

Each dot is one video. Dots that sit close together are videos of someone moving in a similar way.

THE RESULT

The varied quarter beat the random quarter in all 40 of 40 comparisons, with 3.5% lower prediction error.

Why the margin is credible

A 4% margin is small enough to be chance, so we included a selection designed to fail: the same clip count, drawn almost entirely from a handful of people in a handful of rooms.

It lost 39 of 40 comparisons, +6% worse. The measurement can therefore tell a good selection from a poor one, which makes the smaller margins measurements rather than noise.

What this does not show

Training on the full collection remains best. It beats every subset we selected. Choosing well does not substitute for having more.

The narrower claim: at a quarter of the data, a varied selection recovers about 45% of the difference between a random quarter and the full collection — the figure that matters when everything is not an option.

A finding about EgoVerse

A clip is not a consistent unit. Median length is 90 seconds in one collection and 11 in another; 13.9% of the largest collection is a whole session stored as one file.

Matched on clip count, removing those looks harmful (1.7% worse) — but one long clip holds 60× the footage. Matched on footage, removing them is 5% better, 10 comparisons out of 10.

Limitations

The score measures how accurately a model predicts human hand motion, not whether a robot completes the task. The EgoVerse authors use the same proxy for the same reason, and say so in their paper.

The scoring model never sees the video — a consequence of the 84.6 GB figure.

One task, one collection. Transfer is untested.

Tested on 572 clothes-folding clips recorded by 20 people across 16 rooms — the only EgoVerse collection that records who filmed each clip and where, which the held-out comparison requires. 49 removed as unusable. Fu